Hepatitis A virus

Background document for the
WHO Guidelines for drinking-water quality and the
WHO Guidelines on sanitation and health

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Contents

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Highlights

- There is strong evidence linking transmission of Hepatitis A virus (HAV) with unsafe drinking-water, inadequate sanitation and poor hygiene. HAV is an important waterborne pathogen. Many drinking-water outbreaks have been reported.
- Transmission is primarily by the faecal–oral route and infrequently by sexual or parenteral routes. HAV is spread through person-to-person contact. Contaminated food, drinking-water and fomites are also important routes of transmission.
- Humans are the only known reservoir for human-disease causing strains of HAV.
- Infections may be asymptomatic or cause infectious hepatitis (hepatitis A), an acute liver disease characterized by fever, malaise, nausea, anorexia, abdominal discomfort, and eventually jaundice. Hepatitis A is clinically indistinguishable from other forms of hepatitis.
- In low-income settings, there is high endemicity, and children are infected at an early age. In high-income settings, childhood infections are rare, but older children and adults remain susceptible unless vaccinated. Outbreaks are more likely in high-income settings.
- The infectivity of HAV has not been determined, but based on other enteric viruses it is considered to be highly infective.
- HAV cannot be readily detected or cultivated in conventional cell culture systems. Molecular methods are available for detection of HAV in environmental samples.
- In endemic settings, HAV is often detected in surface waters and wastewaters; HAV is one of the most persistent of the enteric viruses.
- Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from HAV. However, as the virus spreads easily from person-to-person other control measures including adequate hygiene are required.
- Reduction of HAV by wastewater and drinking-water treatment varies depending on the process. Although HAV is moderately resistant to chlorine, it can be effective when adequate doses and contact times are applied.
- Due to the smaller size of HAV relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of HAV in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of HAV.

1 General description

Hepatitis A virus (HAV), recently renamed *Hepatovirus A*, is a species of the genus *Hepatovirus* in the family Picornaviridae (ICTV, 2021). Enteroviruses are also members of this family. HAVs share basic structural and morphological features with other members of the family (Cuthbert, 2001). HAV is a non-enveloped, icosahedral, single-stranded RNA virus measuring from 27 nm to 35 nm in diameter (Kotwal & Cannon, 2014). There are six closely related distinguishable genotypes. Genotypes I–III are associated with human infection and are further subdivided into subgenotypes A and B. Genotypes IV–VI are associated with infection in simians (Cristina & Costa-Mattioli, 2007). There is no evidence for cross-species transmission from simians to humans (van der Poel & Rzeżutka, 2017).

2 Human health effects

HAV causes the liver disease hepatitis A, commonly known as infectious hepatitis. It is characterized by a sudden onset of fever, malaise, nausea, anorexia, abdominal discomfort and eventually jaundice (yellowing of the skin and white of the eyes). Itchiness, dark urine and pale stools may also occur. Symptoms of hepatitis A develop, on average, around 4 weeks after infection, although some cases are asymptomatic. Hepatitis A is clinically indistinguishable from other forms of hepatitis. After ingestion, HAV replicates in the alimentary tract and passes to the liver through the bloodstream. It replicates in liver hepatocyte cells to produce virus progeny that are released in bile, passed into the intestine and are excreted in faeces (Cuthbert, 2001).

The damage to liver cells results in the release of liver enzymes such as aspartate aminotransferase and alanine aminotransferase, which are detectable in the bloodstream and used as a diagnostic marker of liver damage. The damage also results in the failure of the liver to remove bilirubin from the bloodstream. This accumulation of bilirubin causes the typical symptoms of jaundice and dark urine. In most cases, hepatitis A resolves without treatment, providing lifelong immunity, but it can sometimes progress from simple jaundice to life threatening acute liver failure (Cuthbert, 2001; Webber, 2005; WHO 2023a).

The course of hepatitis A differs depending on the virus genotype and the age and health status of the host. In general, the severity of illness increases with age and those with pre-existing liver problems. The majority of younger children experience asymptomatic infection or mild symptoms, while older children and adults often experience symptomatic disease. Although mortality is generally less than 1%, repair of the liver damage is a slow process that may keep patients incapacitated for 6 weeks or longer. Mortality as a result of acute liver failure can occur in those over 40 years of age (Cuthbert, 2001).

3 Epidemiology

Disease incidence and transmission rates are associated with low socioeconomic status, low access to safe water and sanitation, as well as person-to-person transmission through sexual contact or injecting drugs using contaminated equipment (Franco et al., 2012; WHO 2023a). In 2019, it was estimated that HAV was responsible for 159 million acute infections, resulting in

39 000 deaths (WHO, 2022a). This burden was not uniformly distributed worldwide. The highest rates of acute infections and deaths occurred in low- and lower-middle-income countries. In such settings, endemicity is high, and children are often infected at a very early age. By contrast, childhood infections are rare in high-income countries, so older children and adults remain susceptible unless vaccinated.

Circulation of HAV genotypes and subtypes varies geographically, with genotype I being most prevalent worldwide and subgenotype IA being more common than IB (Cuthbert, 2001). Subgenotype IIIA circulates widely in some Asian countries and appears to be increasing (Nainan et al., 2006; Yun et al., 2011; Ishii et al., 2012; Lee et al., 2012) and has caused outbreaks in Europe (Stene-Johansen et al., 2005). Genotype II is rarely isolated but is thought to have multiple geographic sources (Desbois et al., 2010).

A safe and effective vaccine is available to prevent hepatitis A (WHO, 2022a). Some countries have introduced routine immunization of children. In contrast, others target those at higher risk of infection, such as those with chronic liver disease, people who may be exposed to hepatitis A through their job or close contact with someone with hepatitis A, travellers to areas where the virus is endemic, drug users, and men who have sex with men (WHO, 2022a).

4 Transmission

HAV is transmitted primarily by the faecal–oral route. Infection can also be spread by parenteral drug taking (Averhoff et al., 2015), blood transfusion (Lefeuvre et al., 2022) and sexual activities (Franco et al., 2012). Person-to-person spread is probably the most common route of transmission in all settings. In endemic areas, most people in settings with household crowding, poor levels of sanitation and inadequate water supplies become infected in early childhood without developing symptoms (Franco et al., 2012). Such infection results in essentially lifelong immunity. In settings with improved water and sanitation, lack of natural exposure to HAV results in populations of susceptible individuals and, thus, the possibility of large outbreaks. Infections occur later in life in such settings and may cause more severe symptoms. Under these conditions, vaccination is an effective strategy to prevent HAV outbreaks. Travelers from low and intermediate endemic regions to high endemic areas without immunization have a high risk of HAV infection. Contaminated food and water are important sources of outbreaks and are associated with infected food handlers, faecally contaminated water used to irrigate produce, faulty wastewater and drinking-water treatment and shellfish harvesting from sewage-contaminated waters (Cuthbert, 2001; van der Poel & Rzeżutka, 2017).

HAV is considered to be highly infective with an estimated $TCID_{50}$ (the number of organisms required to infect 50% of tissue culture cells) of 10 to 100 (Boone & Gerba, 2007), but the precise infective dose sufficient to cause illness in humans is unknown, and infectivity differs among strains. HAV infection has a relatively long incubation period of 28 to 30 days on average (ranging from 2 to 6 weeks) (Koff, 1992). Infected individuals can shed up to 10^9 viral particles/g of faeces (Kotwal & Cannon, 2014). Shedding can begin several weeks before symptoms occur, reaching its peak concurrently with gastrointestinal symptoms and tapering off as the patient shows symptoms of jaundice. Viral particles cannot generally be detected in stool by the time symptoms end (Jacobsen, 2009).

5 Detection methods

Wild-type HAV cannot be readily detected or cultivated in conventional cell culture systems. Therefore, virus identification in environmental samples is based on the use of molecular techniques (Corpuz et al., 2020). They are not only used for the detection but also for genotyping of HAV strains (Nainan et al., 2006). However, distinguishing between viable and non-viable viruses is a shortcoming of molecular methods. Combining cell culture infectivity and molecular methods may make it possible to improve the detection of infectious HAV, including those that do not produce visible cytopathogenic effects. The use of combined methods has been described for HAV, although the method was not used in field studies (Jiang et al., 2004).

6 Source and occurrence

In a systematic review and meta-analysis carried out by Takuissu et al. (2023), the overall prevalence rate of HAV in various water matrices was 16.7%. Prevalence rates in specific matrices are provided in the specific sub-sections below. The prevalence rate was higher in low-income countries (29.0%); high-income countries had a prevalence rate of 10.8%. Africa and Eastern Mediterranean were the regions with higher HAV prevalence rates. Persistence and infectivity have been determined using cell culture-adapted strains (van der Poel & Rzeżutka, 2017). HAV is persistent compared with other picornaviruses, having shown to be stable in laboratory conditions at very low pH levels, temperatures up to 80 °C, and low levels of relative humidity (Scholz et al., 1989; Mbithi et al., 1991; Wang et al., 2015). In a meta-analysis, Bertrand et al. (2012), developed a model to predict the time in days for the first $\log_{10}$ reduction of viruses, including HAV, as a function of temperature and matrix. In simple matrices, such as drinking-water and groundwater stored at 5 °C to 25 °C the time to the first $\log_{10}$ reduction ranged from 2.1 days to 1.6 days. In complex matrices, such as surface waters and sewage stored at 5 °C to 25 °C, the time to the first $\log_{10}$ reduction ranged from 0.7 days to 1.9 days.

6.1 Faecal pollution sources

HAV has been found in high concentrations in raw sewage, treated wastewater and sludge in a number of tropical to semi-tropical countries and countries with temperate climates (van der Poel & Rzeżutka, 2017; Corpuz et al., 2020). For example, HAV was detected in 9.1% to 96% of samples of treated wastewater from a wastewater treatment plant with activated sludge processes with the highest recorded concentration of 9.8×10^8 GC (genome copies)/L. HAV was detected in 25% of samples of treated activated sludge with concentrations ranging from 3.1 to 5.4×10^5 GC/L (Schlindwein et al., 2010). A summary of the presence of HAV (and other viruses) in sewage sludge has been prepared by Gholipour et al. (2022). In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of HAV in samples of untreated (n=2840 from 56 studies) and treated (n=1649 from 34 studies) wastewater were 31.4% and 18%, respectively.

6.2 Surface water

HAV has been found in rivers, lakes and estuaries. In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of HAV in 5851 samples of surface water from 66 studies was 15%. In one study on the Amazon Basin, 92% of water samples tested positive for HAV, with viral loads varying from 60 to 5.5×10^3 GC/L (De Paula et al., 2007). In another study, little decay was observed; HAV was detected for more than 48 days in river water, although persistence is likely to depend on the presence and composition of other organisms in the source (Springthorpe et al.,1993). Therefore, the persistence of HAV in surface water can be considered long.[1]

6.3 Groundwater

In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of HAV in 2165 samples of groundwater from 12 studies was 0.3%. Fongaro et al. (2015) reported that one in five samples of groundwater from deep wells in Brazil contained HAV with a concentration of 4.6×10^5 GC/L. In fresh groundwater, HAV may persist for up to 56 days or more in the absence of microorganisms (Enriquez et al., 1995). In sterile groundwater, HAV survived without reduction in numbers over 8 weeks at 25 °C. However, a 2 $\log_{10}$ reduction was observed in non-sterile groundwater after 2 weeks, suggesting that microbial predation may play a role in inactivation (Sobsey et al., 1986).

6.4 Drinking-water

In a systematic review and meta-analysis (Takuissu et al., 2023), the prevalence rate of HAV in 1388 samples of drinking-water from 11 studies was 0.3%. HAV has been shown to persist for long periods of time at 4 °C with a 1.6 $\log_{10}$ reduction over 56 days in drinking-water while at room temperature, the time taken for 2 $\log_{10}$ reduction was 27 days (Enriquez et al., 1995).[1] HAV persisted in bottled mineral water for at least 1 year at similar temperatures (Biziagos et al., 1988).

6.5 Seawater

In seawater, HAV has shown temperature-dependent reduction, persisting for only 11 to 25 days at 19 °C to 25 °C and persisting for up to 671 days at 4 °C to 5 °C (Fong & Lipp, 2005). Another study reported a 4 $\log_{10}$ reduction in HAV titre in 4 weeks in seawater samples (Callahan et al., 1995).

6.6 Soil and crops

Both soil and crops can be contaminated with enteric viruses through irrigation with contaminated water or the application of organic waste. Cell culture adapted strains of HAV have been shown to survive on foods for several months with little reduction in their infectivity (Kotwal & Cannon, 2014). In soil, it has been shown to remain infective after 12 weeks, including multiple types of soil and those contaminated by sewage effluent (Rodríguez-Lázaro et al., 2012; Rzeżutka & Cook, 2004). However, a 2 $\log_{10}$ inactivation was observed in soil incubated at 25 °C for 8 weeks (Sobsey et al., 1986).

[1] Based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: Detection period for infective stage in water at 20 °C: Short, up to 1 week; Moderate, 1 week to 1 month; Long, over 1 month.

6.7 Surfaces

HAV excreted in faeces can persist for days on non-porous surfaces, with persistence facilitated by the presence of organic matter in faeces (Mbithi et al., 1991). Bae et al. (2012) demonstrated reductions of 1.4 $\log_{10}$ to 2.3 $\log_{10}$ of plaque-forming units on various food contact surfaces over 28 days.

7 Significance of sanitation

Safe sanitation systems are important barriers against the spread of HAV, with many outbreaks associated with sewage contamination (van der Poel & Rzežutka, 2017). However, HAV is highly infective and spreads easily from person-to-person, through contaminated food and surfaces, parenteral drug use and sexual activities. Therefore, additional control measures are needed to effectively prevent transmission.

8 Significance in drinking-water

HAV is an important waterborne pathogen. There is strong evidence linking transmission of HAV with unsafe drinking-water with many reported outbreaks (Bergeisen et al. 1985; Bosch et al., 1991; Tallon et al., 2008). Thirty-two outbreaks of HAV were reported between 1971 and 2010 in the United States of America (USA), of which 23 were associated with untreated groundwater. Reported outbreaks in the USA significantly declined after the implementation of increased regulation of groundwater supplies and vaccination against HAV, and no outbreaks were recorded between 2010 and 2017 (Barrett et al., 2019). Due to surveillance and reporting limitations worldwide, drinking-water associated HAV outbreaks may be significantly underestimated (Barrett et al., 2019).

9 Prevention and management

Systematic approaches that identify and control faecal contamination along the whole sanitation service and drinking-water supply chains are needed to manage microbial risks, including from HAV.

Treatment processes based on size exclusion (e.g. filtration through media or soil in septic tank drainage areas) or settling in primary wastewater treatment processes will be limited for removing HAV and other viruses. HAV, being similar to other enteric viruses, is moderately resistant to oxidizing chemical disinfectants such as chlorine and have low resistance to disinfection by UV light.

9.1 Sanitation systems

Sanitation safety plans (WHO, 2022b) should include control measures to safely collect, treat and discharge or reuse wastewater and faecal sludges. Removal or inactivation of HAV by wastewater treatment processes is variable, although the data are limited (van der Poel & Rzežutka, 2017). Based on available evidence, the reduction of HAV by treatment processes is likely to be lower than the reduction of faecal indicator bacteria such as *E. coli* and enteroviruses (Sidhu & Toze, 2009).

There are no data available on removal of HAV by on-site systems. In centralized sewage treatment, the removal of HAV during primary treatment is negligible. Mixed results have been obtained from assessment of activated sludge performance, with various studies reporting between 0% and 96% of treated wastewater contained HAV (van der Poel & Rzeżutka, 2017). When measured, HAV concentrations in raw sewage and treated wastewater were similar, demonstrating little removal (van der Poel & Rzeżutka, 2017).

HAV have been detected in sewage sludge at concentrations of 10^5 GC/L (Schlindwein et al., 2010). When sludge was inoculated with HAV, treatment with ferric chloride achieved a 1 to 1.5 $\log_{10}$ reduction in HAV over 60 days at 20 °C. In biosolids treated with lime and alum and inoculated with HAV there was an immediate a 3.5 $\log_{10}$ reduction of HAV RNA (Wei et al., 2010). Alkaline stabilization inactivated an inoculum of 6 $\log_{10}$/mL in 2 hrs at 28 °C and in 24 hrs at 4 °C (Katz & Margolin, 2007).

9.2 Drinking-water supplies

Within a water safety plan (WHO, 2012; 2023b), control measures to manage potential risks from HAV should include protection of source waters (surface or groundwater) from contamination by human faecal wastes, application of adequate treatment processes and protection of drinking-water quality during distribution and storage. It is likely that the removal of HAV by conventional drinking-water treatment processes (coagulation, flocculation, sedimentation and filtration) will be similar to that reported for other enteric viruses (Teunis et al., 2009). Moreover, conventional treatment can remove 1.5 $\log_{10}$ of noroviruses (Shin & Sobsey, 2015), which are also non-enveloped RNA viruses of similar size as HAV. HAV is moderately resistant to free chlorine[2], although it can be effective provided adequate doses and contact times are applied. For example, the 2 $\log_{10}$ Ct (product of disinfectant concentration and contact time) value calculated for free chlorine disinfection at 20 °C and pH 6 to 9 for HAV strain HM175 is 1 mg.min/L (US EPA, 1991). HAV is sensitive to UV light; a 2 $\log_{10}$ inactivation was achieved at a dose of 11mJ/cm^2 (Hijnen et al., 2006). Ozone is also effective in inactivating HAV (Shin & Sobsey, 2003).

Due to the smaller size of HAV relative to *E. coli*, resulting in lower removal by treatment processes such as filtration, and their greater environmental persistence, the absence of *E. coli* (or alternatively, thermotolerant coliforms) is not a suitable indicator of the absence of HAV in drinking-water. However, the presence of *E. coli* is a suitable indicator of the possible presence of HAV.

[2] Within pathogen species and groups, there are likely to be variations in resistance, which could be further impacted by characteristics of the water supply and operating conditions. Resistance is based on the criteria established by WHO in the Guidelines for drinking-water quality as follows: 99% inactivation at 20 °C where, generally, Low represents a Ct_{99} of < 1 min.mg/L, Moderate 1–30 min.mg/L and High > 30 min.mg/L (where C = the concentration of free chlorine in mg/L and t = contact time in minutes) under the following conditions: the infective stage is freely suspended in water treated at conventional doses and contact times, and the pH is between 7 and 8.

10 ICD-11 classification

XN41M	Hepatitis virus
XN40D	Hepatitis A virus
1E50	Acute viral hepatitis
1E50.0	Acute hepatitis A

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Annex. Approach to content development and declarations of interest

The background documents were developed drawing from the following resources (i) microbial fact sheets from the fourth edition of the WHO Guidelines for Drinking-Water Quality (ii) pathogen-specific chapters in the Global Water Pathogen Project (GWPP) book "*Sanitation and Disease in the 21st Century",* an update of the seminal book by Feachem et al. 1983 *"Sanitation and Disease: Health Aspects of Excreta and Wastewater Management*" and (iii) a rapid review of the literature at the start of the development of the background documents and before finalization in December 2023, focusing on identifying additional recent high quality literature reviews on a given pathogen. The documents were prepared by members of the WHO drinking-water quality microbiology group as well as external scientists with expertise in the relevant pathogens. The documents went through several iterations based on inputs from members of the WHO drinking-water quality microbiology working group and WHO technical staff. In addition, each document was reviewed by at least two external peer reviewers including authors of the GWPP book chapters. The extensive review process was designed to ensure technical accuracy and to identify any data gaps. The background documents were discussed at three expert meetings to resolve any issues and ensure the content reflected a consensus of expert opinions. Upon satisfactory resolution of any further data gaps or issues, the background documents went through technical editing before finalization.

Before the key expert meetings mentioned above, and for this work more broadly, all microbiology working group members submitted declarations of interest to WHO. These were to disclose any potential conflicts of interest that might affect, or might reasonably be perceived to affect, their objectivity and independence in relation to the subject matter of the guidance. WHO reviewed each of the declarations and concluded that none could give rise to a potential or reasonably perceived conflict of interest related to the subjects discussed at the meetings or covered by the guidance.